

Which form of transport has the smallest carbon footprint?

How can individuals reduce their emissions from transport?

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Update

This article was first published in 2020. It was updated in 2023 with more recent data.

Transport accounts for around one-quarter of global carbon dioxide (CO₂) emissions from energy.¹ In some countries — often richer countries with populations that travel often — transport can be one of the largest segments of an individual's carbon footprint.

If you need to travel locally or abroad, what is the lowest-carbon way to do so?

In this chart, we see the comparison of travel modes by their carbon footprint. These are measured by the amount of greenhouse gases emitted per person *to travel one kilometer*.

This data comes from the UK Government's Department for Energy Security and Net Zero. It's the emission factors companies use to quantify and report their emissions. While the overall rankings of transport modes will probably be the same, there may be differences across countries based on their electricity mix, vehicle stock, and public transport network.

Greenhouse gases are measured in carbon dioxide equivalents (CO₂eq), accounting for non-CO₂ greenhouse gases and the increased warming effects of aviation emissions at high altitudes.²

Walk, bike, or take the train for the lowest footprint

Over short to medium distances, walking or cycling is nearly always the lowest carbon way to travel. While they're not in the chart, the carbon footprint of cycling one kilometer is usually in the range of 16 to 50 grams CO₂eq per km depending on how efficiently you cycle and what you eat.³

Using a bike instead of a car for short trips would reduce travel emissions by around 75%.

Public transport is usually your best option if you can't walk or cycle. Trains are particularly low-carbon ways to travel. Taking a train instead of a car for medium-length distances would reduce emissions by around 80%.⁴ Using a train instead of a domestic flight would reduce your emissions by around 86%.⁵

In fact, if you took the Eurostar in France instead of a short-haul flight, you'd cut your journey's footprint by around 97%.⁶

What if you can't walk or cycle and don't have access to public transport?

If none of the above are options, what can you do?

Driving an electric vehicle (EV) is your best mode of private transport. It emits less than a petrol or diesel car, even in countries with a fairly high-carbon electricity mix. Of course, powering it from a low-carbon grid offers the greatest benefits.

The chart above only considers emissions of EVs during their *use* phase — when you're driving. It doesn't include emissions from car manufacturing. There have been concerns that when we account for the energy needed to produce the battery, an EV is actually worse for

the climate than a petrol car. This is not true — while an EV does have higher emissions during its production, it quickly “pays back” once you start driving it.⁷ The next best is a plug-in hybrid car.

Then, where you take a petrol car or fly depends on the distance. Flying has a higher carbon footprint for journeys less than 1000 kilometers than a medium-sized car. For longer journeys, flying would actually have a slightly lower carbon footprint per kilometer than driving alone over the same distance.

Let’s say you were to drive from Edinburgh to London, a distance of around 500 kilometers. You’d emit nearly 85 kilograms CO₂eq.⁸ If you were to fly, this would be 123 kilograms — an increase of almost one-third.⁹

Some general takeaways on how you can reduce the carbon footprint of travel:

- Walk, cycle, or run when possible — this comes with many other benefits, such as lower local air pollution and better health;
 - Trains are nearly always the winning option over moderate-to-long distances;
 - If travelling internationally, going by train or boat is lower-carbon than flying;
 - Electric vehicles are nearly always lower-carbon than petrol or diesel cars. The reductions are greatest for countries with a cleaner electricity mix;
 - If traveling domestically, driving — even if it’s alone — is usually better than flying;
 - Car-sharing will massively reduce your footprint — it also helps to reduce local air pollution and congestion.
-

Appendix: Why is the carbon footprint per kilometer higher for domestic flights than long-haul flights?

You will notice that domestic flights have higher CO₂ emissions per passenger-kilometer than short-haul international flights, and long-haul flights have even slightly lower emissions. Why is this the case?

In its report on the CO₂ Emissions from Commercial Aviation, the International Council on Clean Transportation provides a nice breakdown of how the carbon intensity (grams CO₂ emitted per passenger kilometer) varies depending on flight distance.¹⁰

This chart, with carbon intensity given as the red line, shows that at very short flight distances (less than 1,000 km), the carbon intensity is very high. It falls with distance until around 1,500 to 2,000 km, then levels out and changes very little with increasing distance.

This is because take-off requires much more energy input than a flight's “cruise” phase. So, for very short flights, this extra fuel needed for take-off is large compared to the more efficient cruise phase of the journey. The ICCT also notes that less fuel-efficient planes are often used for the shortest flights.

ENDNOTES

1. The IEA looks at CO2 emissions from energy production alone — in 2018, it reported 33.5 billion tonnes of energy-related CO2 [hence, transport accounted for 8 billion / 33.5 billion = 24% of energy-related emissions.

2. Aviation creates several complex atmospheric reactions at altitude, such as vapor contrails, creating an enhanced warming effect. In the UK's [Greenhouse gas methodology paper](#), a "multiplier" of 1.9 is applied to aviation emissions to account for this. This is reflected in the CO₂e factors provided in this analysis.

Researchers — David Lee et al. (2020) — estimate that aviation accounts for around 2.5% of global CO₂ emissions but 3.5% of radiative forcing/warming due to these altitude effects.

Lee, D. S., Fahey, D. W., Skowron, A., Allen, M. R., Burkhardt, U., Chen, Q., ... & Gettelman, A. (2020). [The contribution of global aviation to anthropogenic climate forcing for 2000 to 2018](#). *Atmospheric Environment*, 117834.

3. Finding a figure for the carbon footprint of cycling seems like it should be straightforward, but it can vary quite a lot. It depends on several factors: what size you are (bigger people tend to burn more energy cycling), how fit you are (fitter people are more efficient), the type of bike you're pedaling, and what you eat (if you eat a primarily plant-based diet, the emissions are likely to be lower than if you get most of your calories from cheeseburgers and milk). People often also raise the question of whether you actually eat more if you cycle to work rather than drive, i.e., whether those calories are actually 'additional' to your normal diet.

Estimates on the footprint of cycling, therefore, vary. Based on the average European diet, some estimates put this figure at around 16 grams CO₂e per kilometer. In his book "[How bad are bananas: the carbon footprint of everything](#)", Mike Berners-Lee estimates the footprint based on specific food types. He estimates 25 grams CO₂e when powered by bananas, 43 grams CO₂e from cereal and cow's milk, 190 grams CO₂e from bacon, or as high as 310 grams CO₂e if powered exclusively by cheeseburgers.

4. National rail emits around 35 grams per kilometer. The average petrol car emits 170 grams. So the footprint of taking the train is around 20% of taking a car: $[35 / 170 * 100 = 20\%]$.

5. National rail emits around 35 grams per kilometer. A domestic flight emits 246 grams. So the footprint of taking the train is around 14% of a flight: $[35 / 246 * 100 = 14\%]$.
6. Taking the Eurostar emits around 4 grams of CO₂ per passenger kilometer, compared to 154 grams from a short-haul flight. So the footprint of Eurostar is around 4% of a flight: $[4 / 154 * 100 = 3\%]$.
7. The “carbon payback time” for an average driver is around 2 years.
8. An average petrol car emits 170 grams per kilometer. Multiply this by 500, and we get 85,000 grams (85 kilograms).
9. A domestic flight emits 246 grams per kilometer. Multiply this by 500, and we get 123,000 grams (123 kilograms).
10. Graver, B., Zhang, K. & Rutherford, D. (2018). CO₂ emissions from commercial aviation, 2018. International Council on Clean Transportation.

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